

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2022/2023

JULY EXAMINATION

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

MONDAY, 10 JULY 2023

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) The primary focus of Artificial Intelligence (A.I.) is a rational action. An ideal intelligent agent acts to accomplish its goal. In the context of A.I., explain what is an intelligent agent. (3 marks)
- b) Cognitive modelling approach is one way used to develop an intelligent system. Before such a system can be developed, a precise and testable theories of the human mind must first be discovered. Suggest **THREE (3)** ways to understand how human minds work. Briefly explain each of them. (3 + 3 marks)
- c) You are required to design an automated medical diagnosis system. Based on this requirement, answer the following questions.
- (i) List **FOUR (4)** important components of the task environment. (4 marks)
- (ii) Provide an example for each based on the solution for **Question 1 c) (i)**. (4 marks)
- d) Briefly explain **THREE (3)** main differences between a *rational agent* and an *omniscient agent*. (3 marks)
- e) "*Omniscient agents can be created*".
State your agreement with the statement. Justify your answer. (1 + 4 marks)

[Total: 25 marks]

Question 2

- a) Problem formulation is an important phase when designing a problem-solving agent. Briefly explain the **SIX (6)** components of a formulated problem. (12 marks)
- b) **Figure 1** shows interconnected cities in country P where each city is represented by a node, and the distance (in km) between two nodes is represented by an arc connecting them. Assume that you need to analyse all the possible ways and suggest the suitable path to reach the destination, node O from node A. Apply *Breadth-First Search* and *Depth-First Search* to questions below:

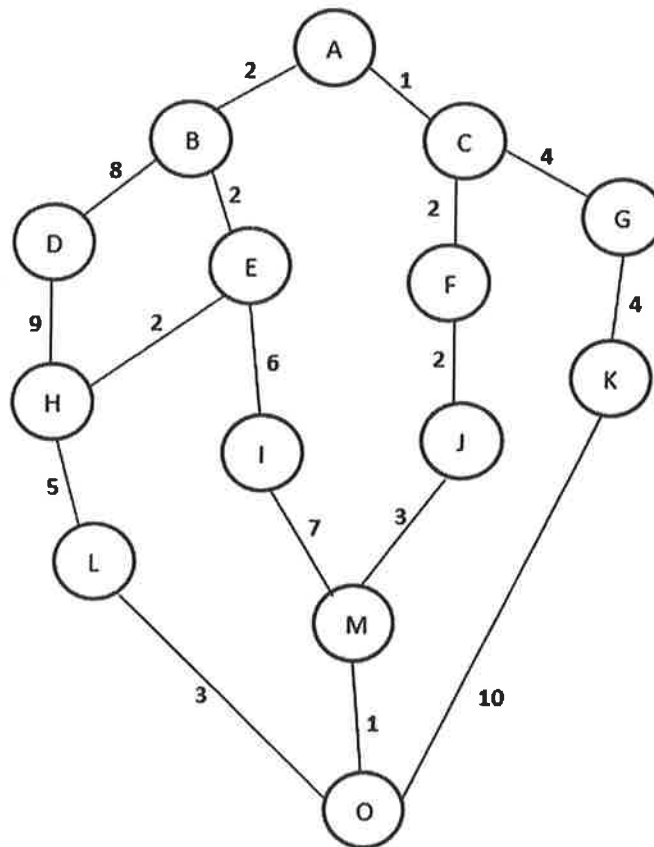


Figure 1 Interconnected cities in country P

Suppose *Breadth-First Search* and *Depth-First Search* are used to solve the problem of searching the shortest path from node A to node O, and both algorithms can deal with repeated states, perform the following tasks for each algorithm.

- Identify the node expansion sequence from the initial state to the goal state in alphabetical order. (2 marks)
- Identify the solution path. (2 marks)
- Calculate the path cost of the solution. Show the working steps. (4 marks)
- Provide a table to interpret the performances of *Breadth-First Search* and *Depth-First Search* in terms of completeness and optimality based on the solution for **Question 2 b)**. (5 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3**

a) A game can be formally defined with six elements.

- (i) Briefly explain **THREE (3)** problem formulations. (6 marks)
- (ii) Calculate the *minimax value* for the nodes (A - F) in the game tree shown in **Figure 2**. (6 marks)

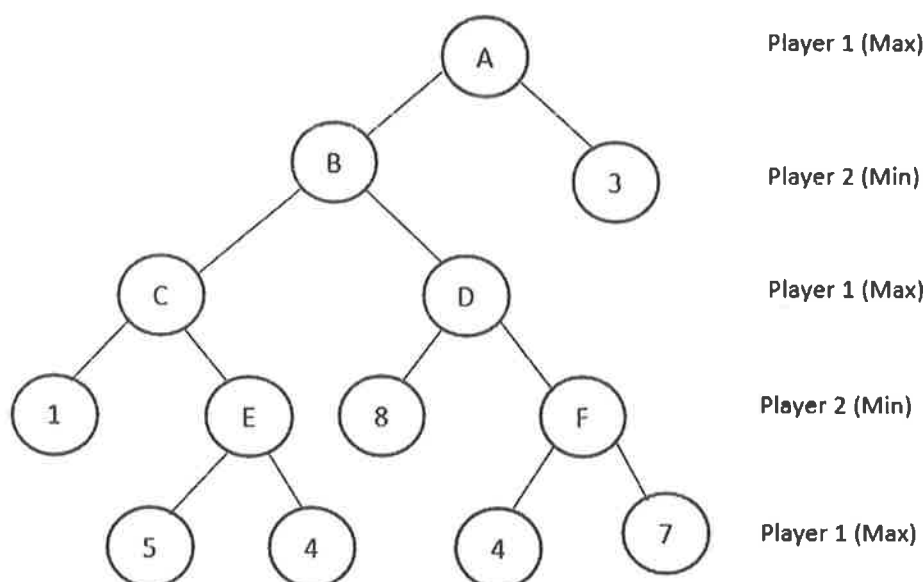


Figure 2 Game tree

- (iii) Identify Player 1's *minimax decision*. To answer this question, you are required to redraw the entire game tree provided in **Figure 2** and highlight the path. Justify the answer. (3 marks)
- (iv) Explain why players *MAX* and *MIN* are named so? (2 marks)

b) A game can be formally defined with the following elements:

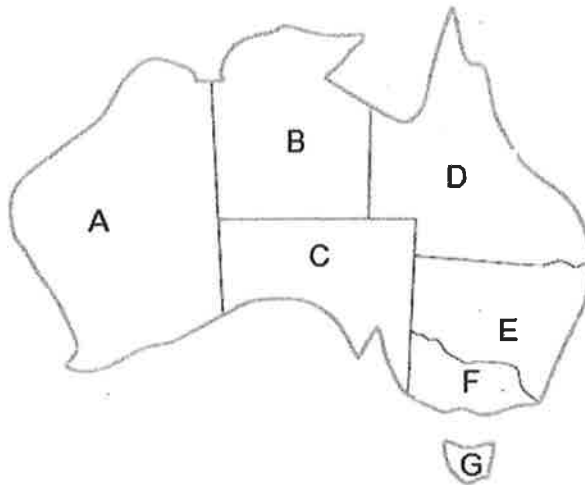
INITIAL STATE, S_0 .
ACTIONS (s)
TO-MOVE (s)
RESULT (s, a)
IS-TERMINAL (s)
UTILITY (s, p)

- (i) Provide a brief description to each of the elements above for a *Tic-Tac-Toe* problem. (7 marks)
- c) *Cut-off search* is a useful technique for balancing the computational complexity of playing game algorithm. However, there are limitations in this search algorithm. Identify **ONE (1)** limitation of *cut-off search*. (1 mark)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4**

- a) Briefly define the *Constraint Satisfaction Problem (CSP)*. (2 marks)
- b) Suppose that, we are given a map of X as in **Figure 3** that shows each of its states and territories. Then, we are given a task of colouring each region with either red, green or blue, in such a way that no neighbouring regions have the same colour.

**Figure 3 Map of X**

- (i) Suppose the colouring task is formulated as an *CSP*, identify its *variables*, *domains* and *constraints*. (6 marks)
- (ii) Based on the answer provided in **Question 4 b) (i)**, identify **ONE (1)** possible solution to this problem. (2 marks)
- (iii) "*A solution to a CSP is a consistent and a complete assignment*". Briefly describe this statement. (2 marks)
- c) Briefly explain **TWO (2)** types of *local consistency*. (5 marks)
- d) Briefly describe *global constraint*. Then, explain why a *global constraint* such as *Alldiff* is still preferred over a set of *binary constraints*. (4 + 2 marks)
- e) Give **ONE (1)** example of game that needs to satisfy an *Alldiff* constraint. Then, briefly describe the game. (1 + 1 marks)

[Total: 25 marks]